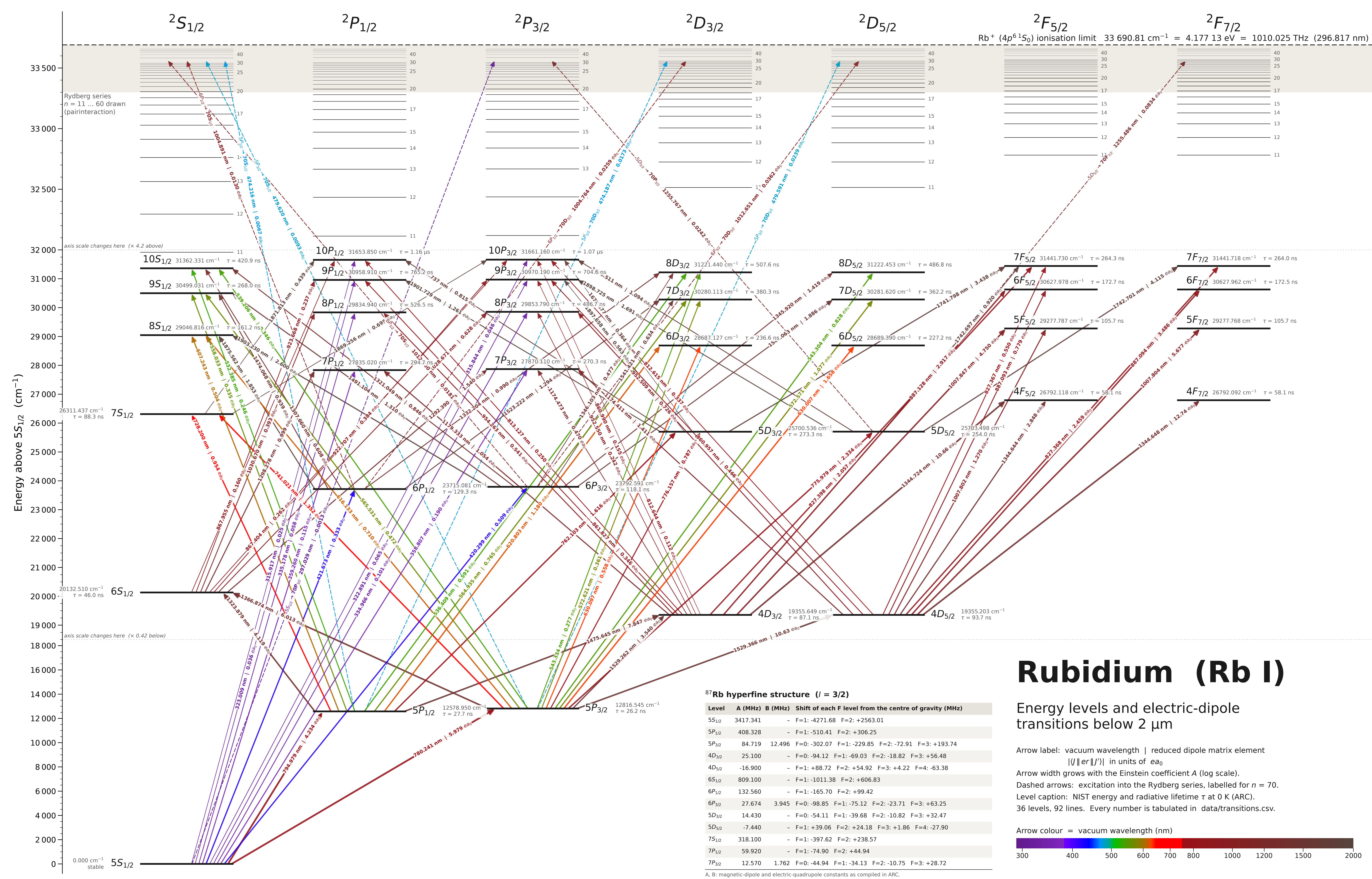
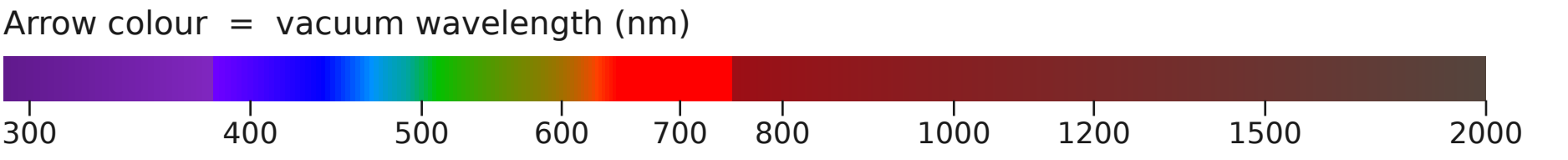




Rubidium (Rb I)

Energy levels and electric-dipole transitions below 2 μm

Arrow label: vacuum wavelength | reduced dipole matrix element
|⟨J||er||J'⟩| in units of ea₀
Arrow width grows with the Einstein coefficient A (log scale).
Dashed arrows: excitation into the Rydberg series, labelled for n = 70.
Level caption: NIST energy and radiative lifetime τ at 0 K (ARC).
36 levels, 92 lines. Every number is tabulated in data/transitions.csv.



Rydberg excitation wavelengths (vacuum, nm)

Lower → series	n = 30	n = 40	n = 50	n = 70	n = 100	n → ∞	d ₇₀ pairint.	d ₇₀ ARC
5S _{1/2} → nP _{3/2}	298.113	297.510	297.247	297.029	296.918	296.817	0.0013	0.0023
5P _{1/2} → nS _{1/2}	477.099	477.510	474.788	474.216	473.927	473.667	0.0067	0.0067
5P _{1/2} → nD _{3/2}	476.682	475.318	474.707	474.187	473.917	473.667	0.0173	0.0166
5P _{3/2} → nS _{1/2}	482.570	480.916	480.205	479.620	479.324	479.059	0.0093	0.0095
5P _{3/2} → nD _{5/2}	482.143	480.747	480.122	479.591	479.315	479.059	0.0239	0.0230
4D _{5/2} → nP _{3/2}	704.765	701.404	699.946	698.736	698.121	697.564	0.0100	0.0092
4D _{5/2} → nF _{7/2}	703.548	700.913	699.701	698.649	698.092	697.564	0.0559	0.0582
6S _{1/2} → nP _{3/2}	745.611	741.850	740.219	738.866	738.178	737.556	0.0055	0.0061
6P _{1/2} → nS _{1/2}	1017.930	1010.598	1007.464	1004.891	1003.596	1002.433	0.0130	0.0130
6P _{1/2} → nD _{3/2}	1016.033	1009.855	1007.100	1004.764	1003.554	1002.433	0.0259	0.0253
6P _{3/2} → nS _{1/2}	1026.025	1018.576	1015.393	1012.780	1011.464	1010.283	0.0181	0.0181
6P _{3/2} → nD _{5/2}	1024.097	1017.821	1015.023	1012.651	1011.421	1010.283	0.0362	0.0354
5D _{5/2} → nP _{3/2}	1275.375	1264.411	1259.681	1255.767	1253.782	1251.986	0.0242	0.0236
5D _{5/2} → nF _{7/2}	1271.394	1262.817	1258.887	1255.486	1253.687	1251.986	0.0834	0.0846
7S _{1/2} → nP _{3/2}	1382.573	1369.698	1364.148	1359.560	1357.233	1355.129	0.0114	0.0118
4F _{7/2} → nD _{5/2}	1478.152	1465.114	1459.322	1454.424	1451.889	1449.545	0.0186	0.0186
7P _{3/2} → nS _{1/2}	1764.032	1742.128	1732.838	1725.240	1721.485	1718.006	0.0291	0.0291
7P _{3/2} → nD _{5/2}	1758.339	1739.920	1731.758	1724.865	1721.300	1718.006	0.0491	0.0486

Rydberg term energies from pairinteraction (quantum defects), lower level from NIST.
d₇₀ = |⟨J||er||J'⟩| to n = 70 in ea₀, from pairinteraction and from ARC. The two disagree by a factor ≈ 2 for 5S → nP (near-cancelling integral): treat that row as an order of magnitude.